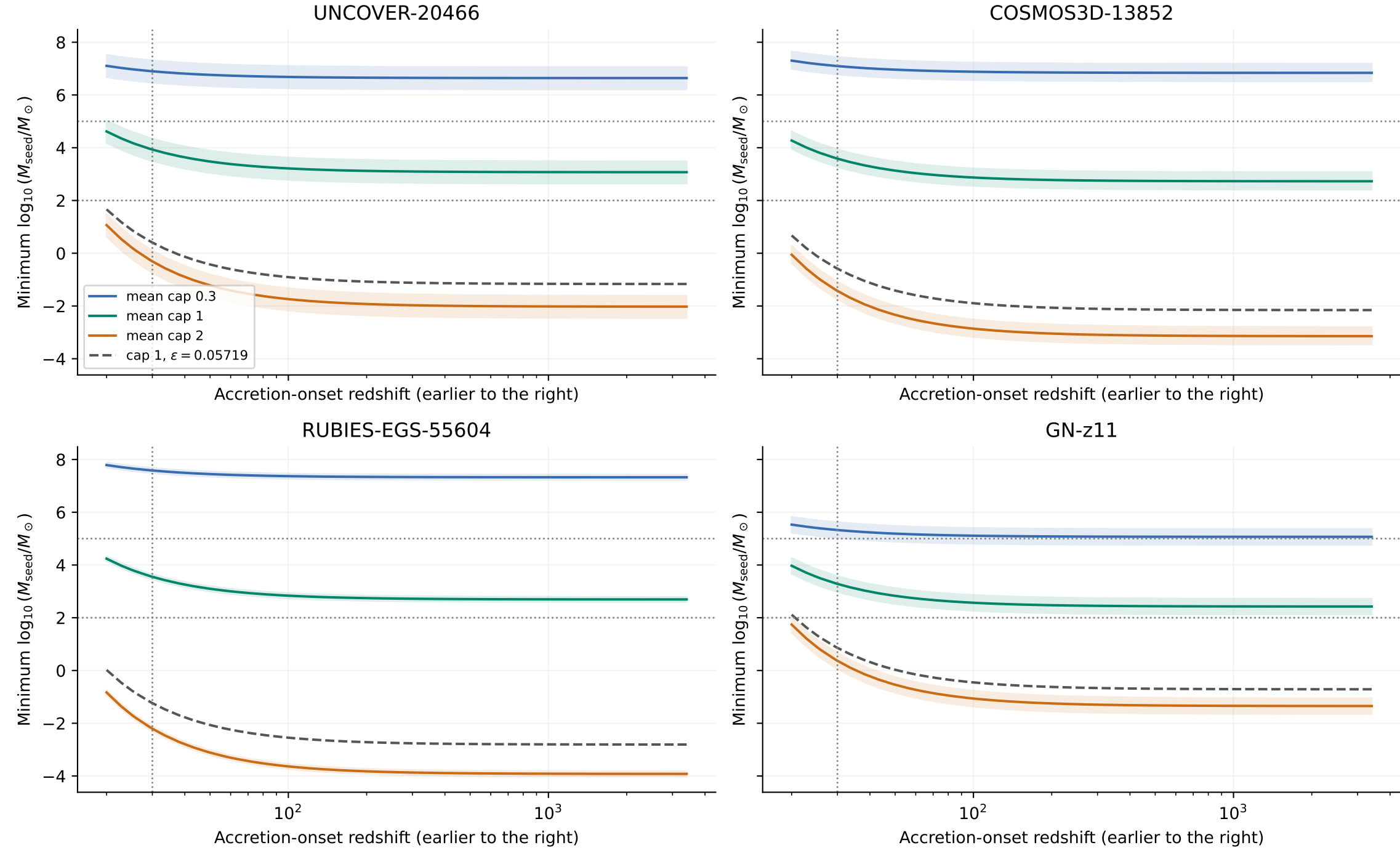


v4: seed requirements under fixed growth caps



Bands: 16-84% of reported log-mass-error draws; no extra systematics. Equality onset is an optimistic formal benchmark.

Colored curves: efficiency 0.1. Dotted lines: onset $z=30$ and seed masses 100 / 100,000 solar masses.

All tracks assume constant efficiency, no mergers, and no gas/feedback limits; PBH viability is not established.